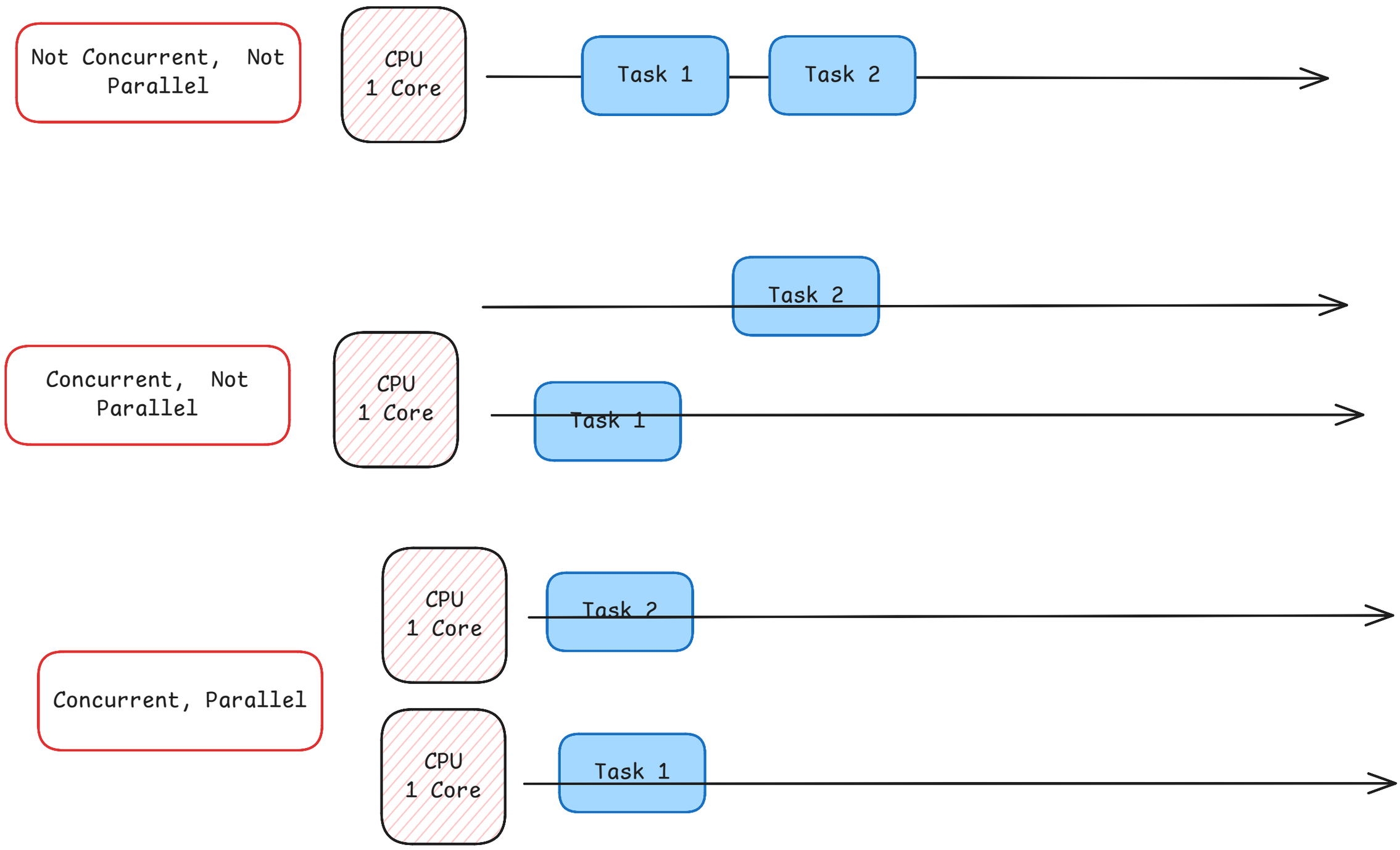
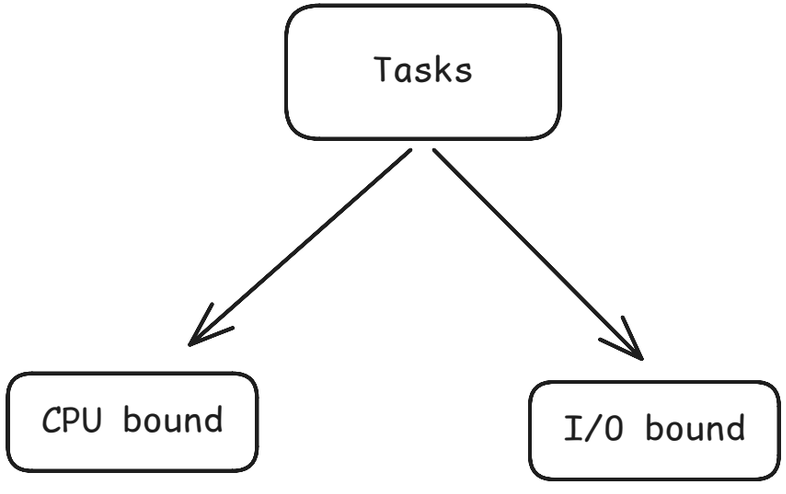
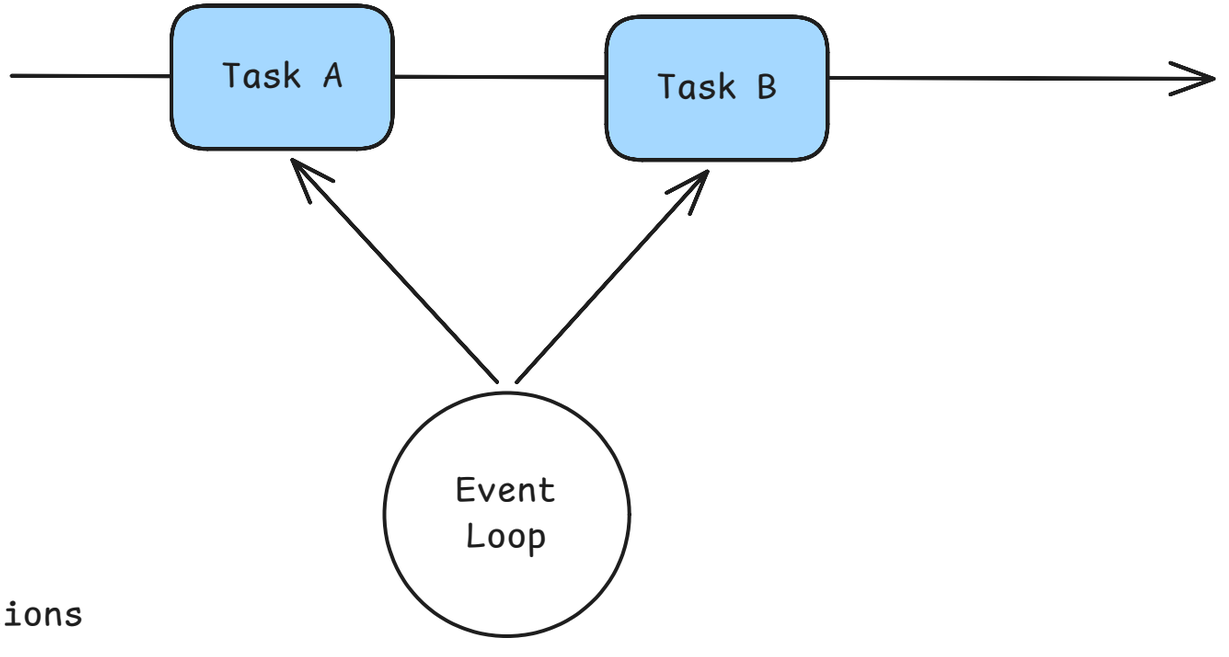
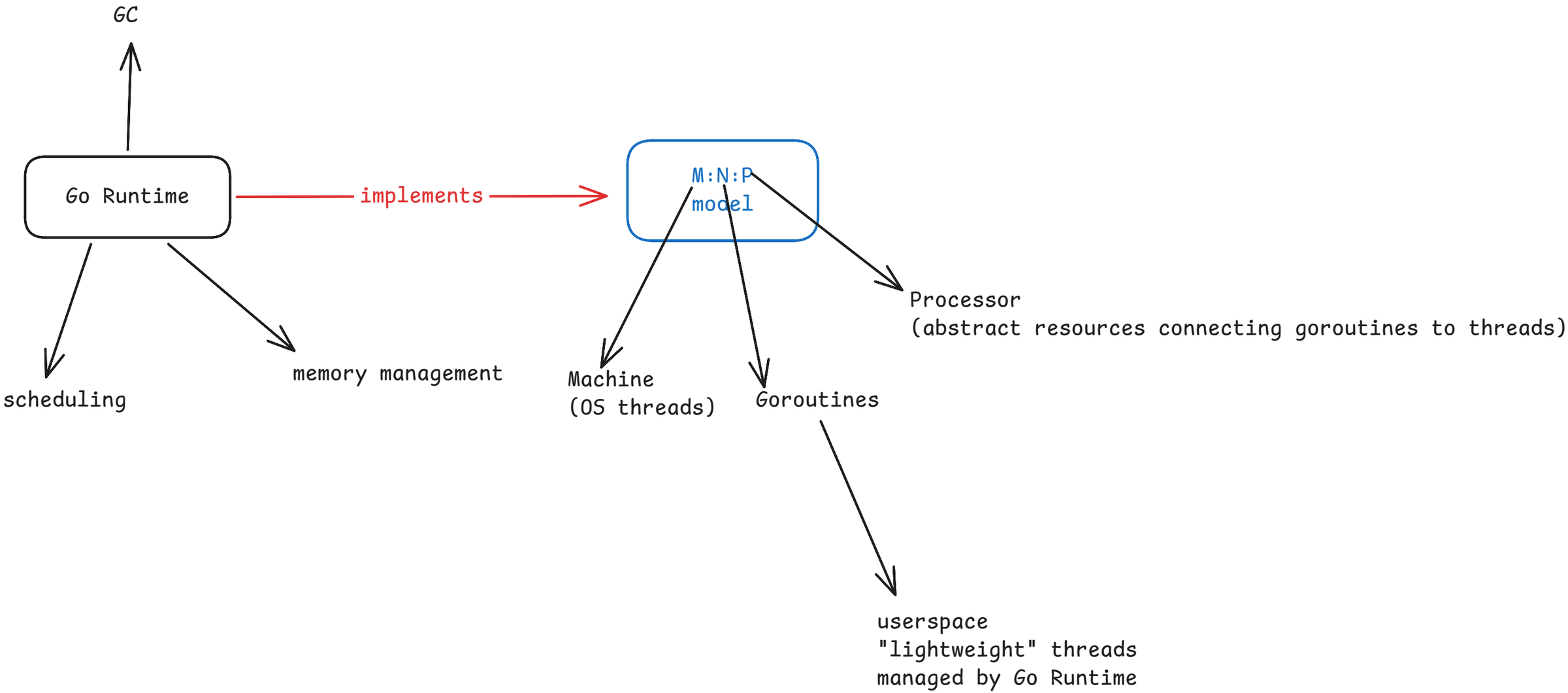
Lecture 09 - Goroutines and the Go Scheduler

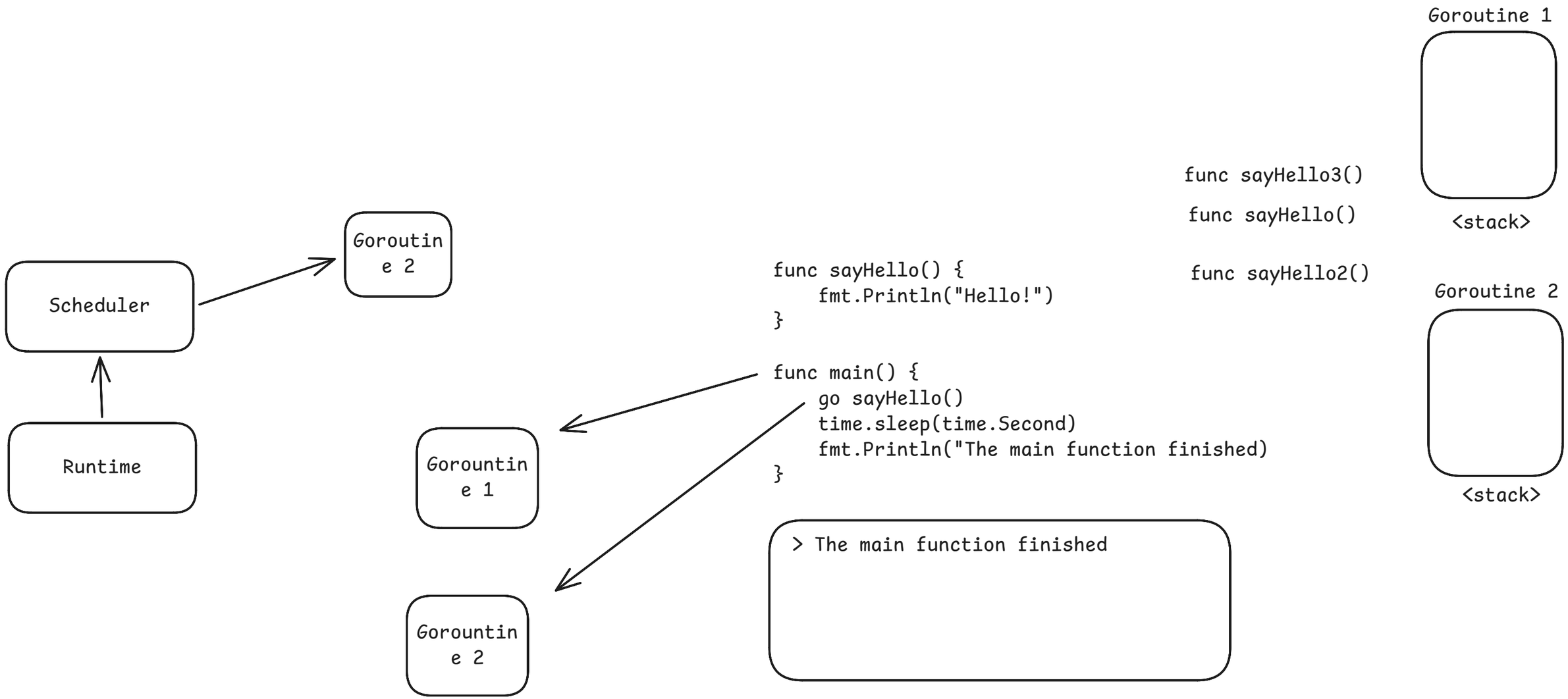

Not Concurrent, Not Parallel
CPU
1 Core
Task 1
Task 2
Concurrent, Not Parallel
CPU
1 Core
Task 2
Task 1
Concurrent, Parallel
CPU
1 Core
Task 2
CPU
1 Core
Task 1

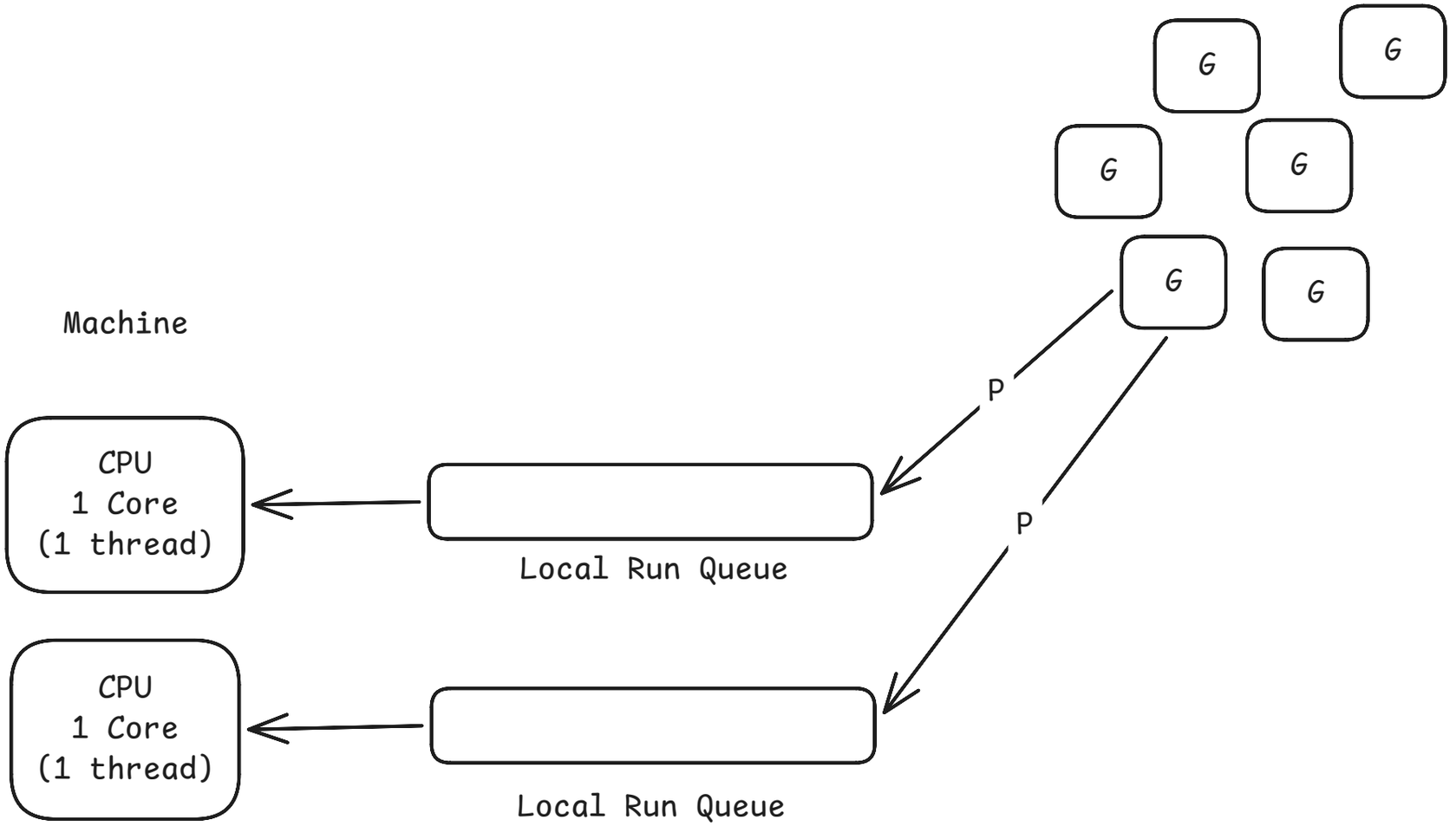


spend much of their
time waiting on external operations




GC
Go Runtime
implements
M:N:P
model
scheduling
memory management
Machine
(OS threads)
Goroutines
Processor
(abstract resources connecting goroutines to threads)
userspace
"lightweight" threads
managed by Go Runtime


Goroutine 1
<stack>
func sayHello3()
func sayHello()
func sayHello2()
Goroutine 2
<stack>
Goroutin e 2
Scheduler
Runtime
Gorountin e 1
Gorountin e 2
func sayHello() {
fmt.Println("Hello!")
}
func main() {
go sayHello()
time.sleep(time.Second)
fmt.Println("The main function finished)
}
> The main function finished



M:N:P concurrency model